

Exercise Sheet 6 for Algorithmen und Datenstrukturen (Sommer 2026)

Hand In: Until 2026-05-30 18:00, on ILIAS.



Problem 1 (Subsequence Sums)

20 points

Solve marburg.kilonova.ro/problems/7 (Subsequence Sums) .

Problem 2

50 points

Derive the correct asymptotic time complexity (Θ -class) of the following method. A formal proof is not required, but you need to show your work.

$f(n)$:

```
// Input:  $n \geq 0$ 
1  for  $i := 1, \dots, n$ 
2      print( $i$ )
3  end for
4  while  $n > 1$ 
5       $n := \lfloor n/2 \rfloor$ 
6       $f(n)$ 
7  end while
```



Problem 3 (Dragon Paths)

40 points

Solve marburg.kilonova.ro/problems/6 (Dragon Paths) .

ILIAS Submission: In your submission to ILIAS, describe your algorithm according to the template,  *Idea*,  *Pseudocode*,  *Correctness*,  *Analysis*.

The pseudocode part can here an informal summary of your kilonova code submission.